P1 习题7 P38

设中点 Q(x,y,z)，定点 P$(-R,0,0)$，$\vec{PQ}=(x+R,y,z)$

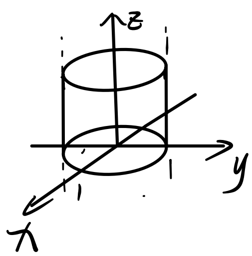
则有 $\vec{PQ}\cdot\vec{OQ}=0$

$x(x+R)+y^2+z^2=0$

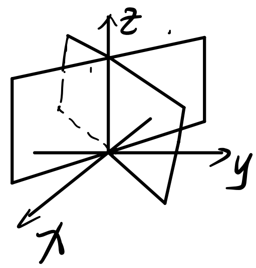
$(x+\frac{R}{2})^2+y^2+z^2=(\frac{R}{2})^2$

P2-5 习题7 P40

I) $x^2+y^2=1$ 圆柱面

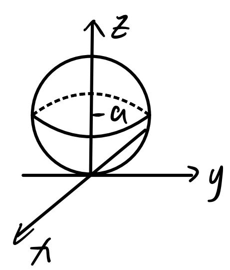


II) $x^2-y^2=0$ 两个过原点相交平面

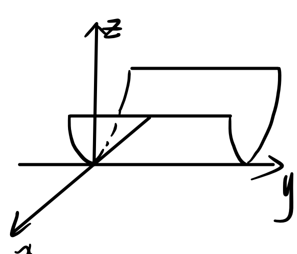


III) $x^2+y^2+z^2=2az$

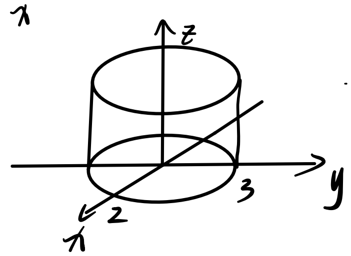
$x^2+y^2+(z-a)^2=a^2$ 以$(0,0,a)$为球心，a为半径的球面



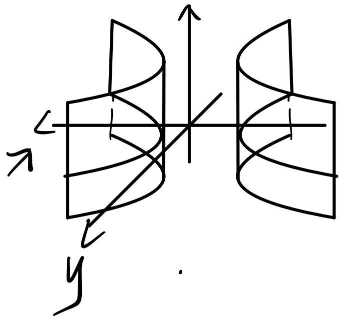
IV) $x^2=2az$ 抛物柱面



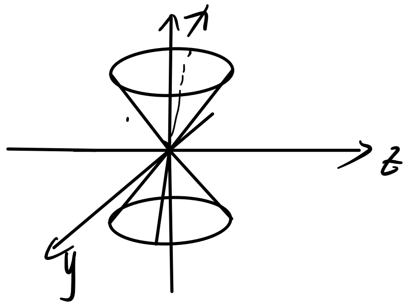
V) $\frac{x^2}{4}+\frac{y^2}{9}=1$ 椭圆柱面



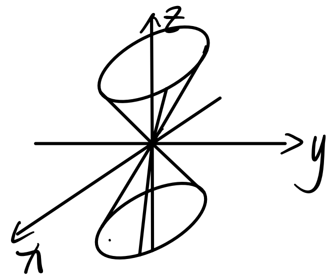
VI) $\frac{x^2}{4}-\frac{y^2}{9}=1$ 双曲柱面



VII) $x^2-y^2=z^2$ 过原点的圆锥面



VIII) $z^2=3x^2+4y^2$ 过原点的椭圆锥面



P6-7 习题7 P41

I) $z_1=z$ $|x_1|=\sqrt{x^2+y^2}$

$z_1=z$ $x_1=\pm\sqrt{x^2+y^2}$

$x^2+y^2+z^2=1$ 球面

II) $x_1=x$ $|y_1|=\sqrt{y^2+z^2}$

$\frac{x^2}{9}+\frac{y^2}{4}+\frac{z^2}{4}=1$ 旋转椭球面

III) $y_1=y$ $|z_1|=\sqrt{z^2+x^2}$

$y^2-x^2-z^2=1$ 旋转双叶双曲面

IV) $x_1=x$ $|z_1|=\sqrt{z^2+y^2}$

$z^2+y^2=5x$ 旋转抛物面

P_8 习题7 P_{43} 2)

$$\begin{cases}(x-1)^2+(y+4)^2+\left(\frac{z}{2}\right)^2=25\\ y+1=0\end{cases}$$

代入 $y=-1$

$$(x-1)^2+\left(\frac{z}{2}\right)^2=16 \quad \text{椭圆}$$

P_9 习题7 P_{44} 2)

$$C:\begin{cases}z=x^2+y^2\\ x+y+z=1\end{cases} \xrightarrow{z=0} C'\begin{cases}\left(x+\frac{1}{2}\right)^2+\left(y+\frac{1}{2}\right)^2=\frac{3}{2}\\ z=0\end{cases} \quad \Sigma:\ \left(x+\frac{1}{2}\right)^2+\left(y+\frac{1}{2}\right)^2\leq\frac{3}{2}$$

P_{10} 习题7 P_{45} 2)

$$xOy:\begin{cases}z^2=2x\\ z=\sqrt{x^2+y^2}\end{cases} \xrightarrow{z=0} C_1\begin{cases}x^2-2x+y^2=0\\ z=0\end{cases} \quad S_1:\ (x-1)^2+y^2\leq 1$$

$$yOz:\begin{cases}z^2=2x\\ z=\sqrt{x^2+y^2}\end{cases} \xrightarrow{x=0} C_2\begin{cases}\frac{z^4}{4}+y^2-z^2=0\\ x=0\end{cases} \quad S_2:\ \left(\frac{z^2}{2}-1\right)^2+y^2\leq 1$$

$$zOx:\begin{cases}z^2=2x\\ z=\sqrt{x^2+y^2}\end{cases} \xrightarrow{y=0} C_3\begin{cases}z^2=2x\\ z=x\\ y=0\end{cases}$$

S_3：抛物线 $z^2=2x$ 与直线 $z=x$ 所围区域 $(x\geq 0)$

P_{11} 习题7 P_{46}

1) $C:\begin{cases}x=\cos t\\ y=2\cos t-1\\ z=3\sin t\end{cases} \quad t\in[0,2\pi]$

$y-2x=-1$, $9x^2+z^2=1$ $\to$ $C:\begin{cases} 2x-y-1=0 \\ 9x^2+z^2-1=0 \end{cases}$

II) $C:\begin{cases} x=t+a \\ y=\sqrt{a^2-t^2} \\ z=\sqrt{2a(a-t)} \end{cases}$ $t\in[-a,a]$

$y^2=2ax-x^2$

$z^2=2a^2-2at=2a(2a-x)$ $\to C:\begin{cases} y-\sqrt{2ax-x^2}=0 \\ z-\sqrt{2a(2a-x)}=0 \end{cases}$

P_{11-14} 习题7 P_{47}

I) $\dfrac{(x-x_0)^2}{a^2}+\dfrac{(y-y_0)^2}{b^2}=1$

$\begin{cases} x=x_0+a\cos t \\ y=y_0+b\sin t \\ z=s \end{cases}$ $t\in[0,2\pi]$, $s\in\mathbb{R}$

II). $\dfrac{y^2}{a^2}-\dfrac{z^2}{b^2}=1$

$\begin{cases} x=s \\ y=a\sec t \\ z=b\tan t \end{cases}$ $t\in[0,2\pi]$, $s\in\mathbb{R}$

III) $-\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2}=1$

令 $\sqrt{\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}}=\sinh t$, $\dfrac{z}{c}=\cosh t$

再令 $\dfrac{x}{a}=\sinh t\cos\theta$, $\dfrac{y}{b}=\sinh t\sin\theta$

$\to\begin{cases} x=a\sinh t\cos\theta \\ y=b\sinh t\sin\theta \\ z=c\cosh t \end{cases}$ $t\in\mathbb{R}$, $\theta\in[0,2\pi]$

IV) $\dfrac{(x-x_0)^2}{a^2}+\dfrac{(y-y_0)^2}{b^2}=z-z_0$

令 $z-z_0=s^2$

$\dfrac{(x-x_0)^2}{(as)^2}+\dfrac{(y-y_0)^2}{(bs)^2}=1$

$$\rightarrow \begin{cases} x = x_0 + as\cos t \\ y = y_0 + bs\sin t \quad t\in[0,2\pi],\ s\in[0,+\infty) \\ z = z_0 + s^2 \end{cases}$$

v) $\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=0$

令 $\frac{z^2}{c^2}=s^2$

$\frac{x^2}{(as)^2}+\frac{y^2}{(bs)^2}=1$

$$\rightarrow \begin{cases} x = as\cos t \\ y = bs\sin t \quad t\in[0,2\pi],\ s\in\mathbb{R} \\ z = cs \end{cases}$$

P_{15} 补充题7 P_7

$$\begin{cases} x+y-3=0 \\ x+z-1=0 \\ x+y+z+1=0 \end{cases} \rightarrow P(5,-2,-4)$$

L过点 $Q_1(1,2,0)$ $\quad \vec{n}=(1,1,1)$

设 Q_1相对π的对称点 $Q_2(x',y',z')$

则有 $\frac{x'-1}{1}=\frac{y'-2}{1}=\frac{z'}{1}=t \rightarrow Q_2(t+1,t+2,t)$

中点 $Q_3\left(\frac{t+2}{2},\frac{t+4}{2},\frac{t}{2}\right)$ 在π上

$$\frac{t+2+t+4+t}{2}+1=0$$

$t=-\frac{8}{3}$ $\quad Q_2\left(-\frac{5}{3},-\frac{2}{3},-\frac{8}{3}\right)$

有 $\vec{s}=(5,-1,-1)$

l: $\frac{x-5}{5}=\frac{y+2}{-1}=\frac{z+4}{-1}$

P_{16-17} 补充题7 P_9

I) $\vec{s_1}=(1,1,2)$ $\quad \vec{s_2}=(1,3,4)$

$P_1(-1,0,1)$ $\quad P_2(0,-1,2)$

$\overrightarrow{P_1P_2} = (1,-1,1)$

$$[\overrightarrow{P_1P_2}, \vec{s_1}, \vec{s_2}] = \begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & 2 \\ 1 & 3 & 4 \end{vmatrix} = 2 \neq 0 \rightarrow \text{异面}$$

II) 设过 $P(0,0,z)$ 直线 l 与 l_1, l_2 相交

$A_1(-1+k, k, 1+2k) \quad A_2(m, -1+3m, 2+4m)$

则有 $\overrightarrow{PA_1} \parallel \overrightarrow{PA_2}$

$$\frac{-1+k}{m} = \frac{k}{-1+3m} = \frac{1+2k-z}{2+4m-z} = t$$

$$\frac{-1+k}{m} = \frac{k}{-1+3m} = t \rightarrow m = \frac{3k-1}{2k-3}, \quad t = \frac{(2k-3)(k-1)}{3k-1}$$

P_{18} 补充题 7 P_{10}

平面束方程 $\pi: x+28y-2z+17+\lambda(5x+8y-z+1)=0$

$$(5\lambda+1)x+(8\lambda+28)y+(-\lambda-2)z+\lambda+17=0$$

$$\vec{n} = (5\lambda+1, 8\lambda+28, -\lambda-2)$$

$$d = \frac{|\lambda+17|}{\sqrt{(5\lambda+1)^2+(8\lambda+28)^2+(\lambda+2)^2}} = 1$$

化简得 $89\lambda^2+428\lambda+500=0$

解得 $\lambda_1 = -2 \quad \lambda_2 = -\frac{250}{89}$

$l_1: \quad 3x-4y-5=0$

$l_2: \quad 387x-164y-24z-421=0$

P_{20} 补充题 7 P_{12}

$z_1 = z \qquad \sqrt{x^2+y^2} = \sqrt{a^2+(\frac{a}{c}z)^2}$

$x^2+y^2 = a^2+(\frac{a}{c}z)^2 \rightarrow \frac{x^2+y^2}{a^2} - \frac{z^2}{c^2} = 1$ 旋转单叶双曲面

思考题

I) 设 $\Pi: z = kx \quad (k>1)$

取过 z 轴的平面 $y = \tan\theta\, x$

$$\begin{cases} y = \tan\theta\, x \\ z = kx \end{cases} \quad \begin{cases} z^2-(x^2+y^2)=1 \\ y = \tan\theta \end{cases} \to P(\rho\cos\theta, \rho\sin\theta, k\rho\cos\theta)$$

$$\rho = \frac{1}{\sqrt{k^2\cos^2\theta - 1}}$$

$$\tan\varphi = \frac{\rho}{z+1} = \frac{\frac{1}{\sqrt{k^2\cos^2\theta-1}}}{\frac{k\cos\theta}{\sqrt{k^2\cos^2\theta-1}}+1} = \frac{1}{k\cos\theta+\sqrt{k^2\cos^2\theta-1}}$$

设 $C(r,\theta,\phi)$

$$\cos\phi = \frac{k\cos\theta+\sqrt{k^2\cos^2\theta-1}}{\sqrt{2k\cos\theta(k\cos\theta+\sqrt{k^2\cos^2\theta-1})}} = \sqrt{\frac{k\cos\theta+\sqrt{k^2\cos^2\theta-1}}{2k\cos\theta}}$$

$$\sin\phi = \frac{1}{\sqrt{2k\cos\theta(k\cos\theta+\sqrt{k^2\cos^2\theta-1})}}$$

$$x_c = \sin\phi\cos\theta = \frac{\cos\theta}{\sqrt{2k\cos\theta(k\cos\theta+\sqrt{k^2\cos^2\theta-1})}} \qquad z_c = \cos\phi$$

$$2k x_c z_c = k\sin2\phi\cos\theta$$

$$(1-z_c^2) - 2k x_c z_c - 2k x_c + (z+1)^2 = 0$$

$$1+z_c - k x_c(z_c+1) = 0$$

$$(z_c+1)(1-kx_c) = 0 \to x_c = 1$$

圆心 $(\frac{1}{k}, 0, 0)$ → 在平面 $z=0$ 上

II) C': $\rho = k\cos\theta - \sqrt{k^2\cos^2\theta-1}$

$\theta=0$ 时, $\rho = k\pm\sqrt{k^2-1}$ → $A(k+\sqrt{k^2-1}, 0)$ $B(k-\sqrt{k^2-1}, 0)$

$\theta = \arccos\frac{1}{k}$ 时 $\rho=1$ → $D(\frac{1}{k}, \sqrt{1-\frac{1}{k^2}})$ → 圆心 $(k,0)$, $R = \sqrt{k^2-1}$

极坐标: $\rho^2-2k\rho\cos\theta+1=0$　步骤1: $\rho=k\cos\theta-\sqrt{k^2\cos^2\theta-1}$

$$\rho^2-2k\rho\cos\theta+1=0 \longrightarrow \text{相同}$$

故 C' 也是个圆弧

$d=k$　　$R^2+r_0^2=k^2-1+1=k^2=d^2 \longrightarrow$ 正交

故 C' 与单位圆垂直相交.